Class-Imbalance Mitigation in Computational Pathology: Methods for Patch Classification and WSI-Level MIL

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July 14, 2026

Abstract

Class imbalance in computational pathology depends on the input regime: labeled image patches support ordinary classification, whereas whole-slide image (WSI) diagnosis is commonly modeled as weakly supervised learning over patch-feature bags. This report is the authoritative method reference for the class-imbalance benchmark. It defines sampling, loss, prior-correction, decoupled-classifier, set-learning, feature-mixing, contrastive-MIL, and dual-expert interventions in the input form each consumes, with cross-entropy as the no-mitigation reference. It also distinguishes faithful source implementations from deliberately adapted variants and states the output semantics and assumptions needed to interpret each intervention.

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1 Introduction

Class imbalance changes what a pathology classifier is rewarded for learning. When common tumor types dominate training data, high aggregate accuracy can coexist with unreliable rare-class recall, especially if evaluation emphasizes prevalence-sensitive metrics. Long-tailed learning and medical-image evaluation literature therefore motivate classwise recall, macro F1, balanced accuracy, and calibration-aware probability metrics as primary outcomes in imbalanced settings (Saini and Susan, 2023; Zhang et al., 2023; Mosquera et al., 2024).

Imbalance methods often assume a specific prediction unit. Image-level methods use labeled images, image embeddings, or directly generated image samples, whereas WSI-level methods use weak labels over bags of instances together with attention, instance ranking, or bag-level representation learning. Each method family is therefore described in the input form it is designed to consume. The broader computational-pathology pipeline these regimes sit in—whole-slide imaging, patching into tiles, and weakly supervised multiple-instance learning over frozen foundation-model features—is introduced in the companion datasets report (Queisler, 2026).

This report defines the class-imbalance mitigation methods themselves, organized by the level at which they intervene and stated in enough mathematical detail to make the imbalance mechanism explicit. The aim is not to introduce a new mitigation method, but to fix the method definitions, assumptions, and fidelity qualifications used when instantiating these methods on specific pathology datasets. The companion experimental protocol defines the controlled datasets, validation grids, evaluation endpoints, calibration procedure, aggregation units, and statistical analysis.

Prediction regimes. Two prediction regimes recur throughout the benchmarks. In the *patch* regime, each labeled crop is an independent example: a frozen foundation-model encoder such as Virchow2 embeds the patch once, and a lightweight classifier head is trained on the resulting embedding (Zimmermann et al., 2024; Vorontsov et al., 2024). In the *WSI-bag* regime, a slide is represented as a bag of frozen patch-feature instances that share one weak slide-level label, and an attention-based MIL head aggregates the instances into a slide prediction. Both regimes consume embeddings from the same pathology foundation-model family but differ in supervision unit. Freezing the encoder isolates imbalance interventions from end-to-end backbone learning. Each non-baseline method exposes one imbalance-specific control; the concrete validation grids are reported in the companion experimental protocol.

2 Methods

The methods are organized by the level at which they intervene in the imbalanced learning problem. Cross-entropy (CE) is treated as the no-mitigation reference condition: it uses the observed training distribution as given and does not rebalance samples, labels, costs, or representations. The remaining methods are grouped as data-level sampling and augmentation, algorithm-level losses and prior corrections, hybrid sampling–loss methods, and representation or architecture-level methods (Saini and Susan, 2023; Zhang et al., 2023).

The comparison is regime-specific. Patch methods consume frozen Virchow2 embeddings of controlled patches and optimize one shared MLP classifier head; CE-based patch methods use the same linear output layer, while CFAL replaces that layer with class prototypes and affinity-based prediction. WSI-bag methods consume capped frozen Virchow2 feature bags and optimize one shared attention-MIL head. Both regimes therefore rely on the same pathology foundation-model family, but they differ in supervision unit: one embedding per selected patch versus aggregation over a bounded set of instances on a slide.

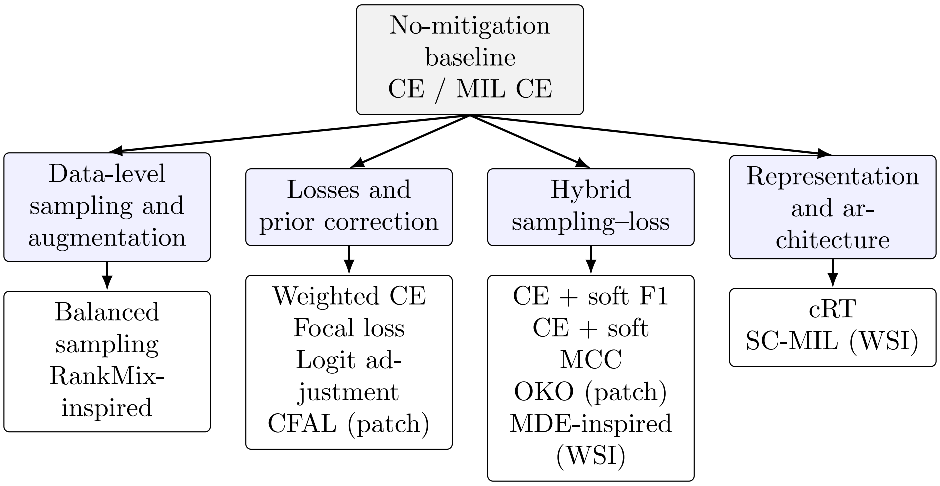

Figure 1: Method organization used in the benchmarks. Cross-entropy is the no-mitigation reference; alternatives intervene through sampling, objectives, prior correction, feature mixing, or representation and classifier design.

2.1 Notation

Let C be the number of cancer-type classes and let n_c denote the number of training examples from class c. For a labeled patch or slide-level bag with label $y \in \{1, \ldots, C\}$, the model outputs logits $\mathbf{z} \in \mathbb{R}^C$ and class probabilities

$$p_c = \frac{\exp(z_c)}{\sum_{j=1}^{C} \exp(z_j)}. \tag{1}$$

For WSI-bag methods, a slide is represented as a bag $B = \{\mathbf{x}_1, \ldots, \mathbf{x}_m\}$ of frozen patch-feature instances. The attention-MIL backbone maps each bag to a bag embedding $\mathbf{b}(B)$ and class logits. Unless a method explicitly changes the sampler, objective, or representation loss, the empirical class frequencies in the training set define the optimization signal.

2.2 Cross-Entropy Baseline

Unweighted cross-entropy is the baseline because it is the standard supervised objective without an explicit class-imbalance correction. For an example with label y, the loss is

$$\mathcal{L}_{\mathrm{CE}}(\mathbf{z}, y) = -\log p_y. \tag{2}$$

In the patch regime, this objective trains the shared linear head on frozen Virchow2 patch embeddings from the controlled manifest. In the WSI-bag regime, the same objective trains the shared attention-MIL model on slide-level feature bags. In both cases, CE uses the observed class distribution directly: majority classes appear more often in minibatches and therefore contribute more loss terms over an epoch. This makes CE an appropriate reference for asking whether an imbalance-specific intervention actually improves the rare-class tradeoff.

2.3 Data-Level Sampling and Augmentation

Data-level methods change what the optimizer sees, rather than changing the mathematical penalty assigned to one observed example. In the taxonomy, this family includes resampling, augmentation, and WSI-specific feature mixing strategies that make minority or informative examples more visible during training (Saini and Susan, 2023).

2.3.1 Balanced Sampling

Balanced sampling changes minibatch composition by sampling training examples with probabilities inversely related to their class support. For an example i with class y_i, the sampler weight is proportional to

$$q_i \propto \frac{1}{n_{y_i}}. \tag{3}$$

After normalization across the training set, this makes the expected class exposure closer to uniform even though the underlying dataset remains imbalanced. The patch variant applies this sampler to patch examples and still optimizes CE. The MIL variant applies the same idea at slide-bag level and also optimizes CE. Balanced sampling therefore tests a pure data-level intervention: it asks whether equalizing training exposure is enough without changing the loss formula.

2.3.2 RankMix-Inspired Feature Mixing

RankMix is a WSI-native data-augmentation method for weakly supervised slide classification. Ordinary image mixup assumes two inputs with compatible spatial shapes, but WSI bags can contain different numbers of feature instances and lack instance-level labels. RankMix addresses this by mixing representative feature subsequences rather than raw slides (Chen and Lu, 2023). The benchmark variant is called *RankMix-inspired* because it preserves the source method's teacher ranking, representative subsequence selection, feature interpolation, mixed labels, and reinitialized student, but does not reproduce the paper's complete combination of instance-max and bag-level objectives. Its student instead uses soft-label CE on the mixed bag prediction. Results therefore test this specified adaptation, not a faithful reproduction of the full source method.

The tested implementation follows the defining two-stage teacher–student procedure. First, a teacher attention-MIL model is trained with slide labels for 30 epochs using the same MIL architecture and optimization settings as plain MIL CE. For each bag $B = \{\mathbf{x}_1, \ldots, \mathbf{x}_m\}$ and target class, the teacher assigns ranking scores to instances using its softmax probability for that class; the top-k instances are kept as a representative subsequence, where $k = \min(|B_a|, |B_b|)$ for the two bags being mixed, and their original within-bag order is preserved after selection. The student MIL model is then reinitialized and trained on bags built from these ranked subsequences.

Mixing itself follows the mixup idea in feature space. For two bags a and b, let H_a and H_b denote the selected instance-feature matrices and $\mathbf{y}_a, \mathbf{y}_b$ the corresponding one-hot slide labels. A scalar mixing coefficient $\lambda \in (0, 1)$ controls how much of each bag is retained:

$$\widetilde{H} = \lambda H_a + (1 - \lambda) H_b, \qquad \widetilde{\mathbf{y}} = \lambda \mathbf{y}_a + (1 - \lambda)\mathbf{y}_b. \tag{4}$$

When λ is close to one, the mixed bag resembles bag a; when it is close to zero, it resembles bag b. Values near $\lambda = \frac{1}{2}$ interpolate more evenly between the two sources. RankMix does not fix λ to a single value. Instead, each mixed example draws

$$\lambda \sim \mathrm{Beta}(\alpha, \alpha), \tag{5}$$

independently. The Beta distribution is a standard choice on the open interval $(0, 1)$ because it is flexible and symmetric when both shape parameters are equal. With density

$$f(\lambda; \alpha, \alpha) \propto \lambda^{\alpha-1}(1 - \lambda)^{\alpha-1}, \qquad \lambda \in (0, 1), \tag{6}$$

the parameter $\alpha > 0$ controls how strongly mixing concentrates around the endpoints versus the center. For $\alpha = 1$, $\mathrm{Beta}(1, 1)$ is uniform on $(0, 1)$. Larger α pushes λ toward $\frac{1}{2}$; smaller α keeps mixtures closer to one source bag. The reference default is $\alpha = 1.0$, and α is selected on validation. The student is trained with soft-label CE on the mixed bags and labels $\widetilde{\mathbf{y}}$. This is a data-level feature-space augmentation while retaining the common attention-MIL prediction architecture.

2.4 Algorithm-Level Losses

Algorithm-level methods keep the training examples fixed but alter how errors are penalized. They are cost-sensitive or difficulty-sensitive objectives: minority examples may receive larger explicit weights, or easy examples may contribute less to the gradient (Saini and Susan, 2023; Scholz et al., 2024).

2.4.1 Class-Weighted Cross-Entropy

Class-weighted CE multiplies each example's CE term by a class-dependent cost:

$$\mathcal{L}_{\mathrm{WCE}}(\mathbf{z}, y) = -w_y \log p_y, \qquad w_c \propto \frac{1}{n_c}. \tag{7}$$

The implementation normalizes the inverse-frequency weights so that the average scale of the loss remains comparable across methods. This method leaves the sample order and model architecture unchanged, but increases the loss contribution of classes with fewer training examples. It is evaluated for both patch classification and attention-MIL.

2.4.2 Focal Loss

Focal loss reduces the contribution of confidently classified examples and concentrates optimization on examples that remain difficult. With focusing parameter γ, the multiclass focal objective is

$$\mathcal{L}_{\mathrm{focal}}(\mathbf{z}, y) = -(1 - p_y)^{\gamma} \log p_y. \tag{8}$$

The reference default is $\gamma = 2.0$ for both regimes; γ is the imbalance control selected on validation while the rest of the training protocol is held fixed. The method is relevant to imbalance because many minority-class examples can remain hard under a classifier dominated by head-class gradients, but the focal term does not directly encode the class counts. It is therefore a difficulty-sensitive loss rather than a sampler or explicit prior correction.

2.4.3 Logit Adjustment

Logit adjustment explicitly corrects for the empirical training prior (Menon et al., 2021). Let

$$\widehat{\pi}_c = \frac{n_c}{\sum_{j=1}^{C} n_j} \tag{9}$$

denote the class prior estimated only from the current training condition. The control $\tau \geq 0$ determines the strength of the correction; $\tau = 0$ recovers ordinary CE. In the train-time form, adjusted logits are used inside the loss,

$$\mathcal{L}_{\mathrm{LA-train}}(\mathbf{z}, y) = -\log \frac{\exp(z_y + \tau \log \widehat{\pi}_y)}{\sum_{c=1}^{C} \exp(z_c + \tau \log \widehat{\pi}_c)}. \tag{10}$$

The positive prior term makes head classes harder to fit during training, inducing a classifier whose unadjusted logits favor a class-balanced target. In the post-hoc form, a model is trained with ordinary CE and corrected only at inference,

$$p_c^{\mathrm{LA}}(\mathbf{x}) = \frac{\exp(z_c(\mathbf{x}) - \tau \log \widehat{\pi}_c)}{\sum_{j=1}^{C} \exp(z_j(\mathbf{x}) - \tau \log \widehat{\pi}_j)}. \tag{11}$$

The sign difference follows from where the correction is applied: the train-time loss learns against prior-shifted logits, whereas the post-hoc rule removes the observed training prior from ordinary CE logits. Both patch and WSI regimes use the same definition at their respective prediction unit. Because either form deliberately changes the implied target prior, its probabilities should not be interpreted as calibrated for the natural test prevalence without an explicit target-prior assumption. Discrimination and calibration are therefore reported together, with post-hoc temperature scaling kept conceptually separate from prior correction.

Balanced Softmax is closely related prior-correction literature: it modifies the softmax denominator using class frequencies and motivates the same distinction between representation learning under a long-tailed source and decision rules for a balanced target (Ren et al., 2020). It is not an additional benchmark method; it is included to situate logit adjustment among count-aware softmax objectives.

2.4.4 Center-Focused Affinity Loss (CFAL)

Center-focused affinity loss (CFAL) replaces the linear softmax head with learnable class prototypes and an affinity-based training objective (Mahbub et al., 2024). The motivation is that minority histology classes can remain visually similar in embedding space even when a frozen encoder already separates them coarsely; pulling examples toward compact class centers and penalizing confusion with nearby wrong-class prototypes can therefore improve tail-class boundaries without changing the sampler.

The tested patch implementation keeps the shared MLP trunk that maps a frozen Virchow2 embedding $\mathbf{x}$ to a hidden representation $\mathbf{z} = f(\mathbf{x}) \in \mathbb{R}^{512}$, but replaces the final linear map with one learnable prototype $\mathbf{w}_c \in \mathbb{R}^{512}$ per class. Both embeddings and prototypes are L2-normalized before affinity computation. The Gaussian affinity between example i and class c is

$$a_{ic} = \exp\left(-\frac{\|\hat{\mathbf{z}}_i - \hat{\mathbf{w}}_c\|_2^2}{\sigma}\right), \qquad \hat{\mathbf{v}} = \frac{\mathbf{v}}{\|\mathbf{v}\|_2}, \tag{12}$$

where σ controls how quickly affinity decays with prototype distance. At inference, class scores are $\log a_{ic}$ followed by softmax normalization. The affinities $a_{ic} \in (0, 1]$ are geometric similarity scores, not class probabilities: they are trained with a margin objective and never optimized to satisfy likelihood or calibration constraints. Applying softmax to $\log a_{ic}$ therefore yields a convenient ranking score for argmax decisions, but it should not be read as a calibrated estimate of $P(y = c \mid \mathbf{x})$ without additional post-hoc mapping.

Training optimizes a margin term on affinities rather than cross-entropy on logits. For batch index i with label y_i, define the per-example margin loss

$$\ell_i = \sum_{c \neq y_i} \max\bigl(0,\, \lambda + a_{ic} - a_{i,y_i}\bigr), \tag{13}$$

which penalizes wrong-class affinities that exceed the true-class affinity by less than margin λ. A diversity regularizer $R(\mathbf{W})$ encourages prototypes to spread apart by penalizing deviations of pairwise squared distances from their mean. The total objective combines class reweighting, a center-focused local weight, and prototype diversity:

$$\mathcal{L}_{\text{CFAL}} = \frac{1}{B} \sum_{i=1}^{B} \frac{1}{E_{y_i}} (1 - a_{i,y_i})^{\gamma}\, \ell_i + R(\mathbf{W}), \tag{14}$$

where E_c is the effective number of training examples in class c derived from class count n_c and hyperparameter β, and γ up-weights examples whose true-class affinity remains low. Unlike the CE plus soft-F1 and soft-MCC hybrids, CFAL does not use balanced oversampling: it is categorized as an algorithm-level patch method because it changes the head and loss while keeping the natural training distribution.

The reference defaults are $\lambda = 0.1$, $\beta = 0.999$, and $\gamma = 2.0$ following Mahbub et al.; all three are held fixed and the affinity bandwidth σ is the imbalance control selected on validation. The paper's $\sigma = 130$ was calibrated for un-normalized AlexNet FC features; on L2-normalized 2,560-dimensional Virchow2 vectors that scale is far too large and collapses all pairwise affinities, so the benchmark evaluates a range matched to the normalized embedding geometry. CFAL is evaluated only in the patch regime because the source method assumes instance-level labels rather than weak slide-level bags.

2.5 Hybrid Sampling–Loss Methods

Hybrid methods combine data-level and algorithm-level changes. The Scholz-style patch objectives below adapt the metric-derived losses of Scholz et al. for imbalanced medical classification (Scholz et al., 2024). Both tested variants use class-balanced oversampling together with metric-derived losses, so their effect cannot be attributed only to a new loss or only to a new sampler.

2.5.1 Scholz-Style CE + Soft F1

The intuition is that CE optimizes the log-probability of the correct class one example at a time, whereas macro F1 evaluates a different object: the balance between precision and recall at class level. Under class imbalance, this distinction matters because a model can reduce CE substantially by fitting frequent classes while still leaving tail-class recall or precision weak.

The obstacle is that ordinary F1 is computed from hard counts after prediction, so it is not directly usable as a gradient-based training loss. The Scholz-style soft-F1 objective replaces those hard counts with probability-weighted counts inside each minibatch. For one class, a confident correct prediction contributes almost one soft true positive, a confident prediction for the wrong class contributes to soft false positives, and low probability on a true class contributes to soft false negatives. Formally, for one-hot labels y_i and predicted probabilities $\hat{y}_i$:

$$\mathrm{TP} = \sum_i y_i \hat{y}_i, \quad \mathrm{FP} = \sum_i (1 - y_i)\hat{y}_i, \quad \mathrm{FN} = \sum_i y_i (1 - \hat{y}_i). \tag{15}$$

These terms keep the familiar meaning of a confusion matrix but make every component differentiable with respect to the model probabilities. They then define soft precision, soft recall, and soft F1:

$$\mathrm{F1}_{\mathrm{soft}} = \frac{2\,\mathrm{precision}_{\mathrm{soft}}\,\mathrm{recall}_{\mathrm{soft}}}{\mathrm{precision}_{\mathrm{soft}} + \mathrm{recall}_{\mathrm{soft}}}. \tag{16}$$

The multiclass implementation applies this construction separately to each class and averages the resulting classwise terms. This macro-style averaging is the imbalance-aware part: each class contributes one term regardless of how many samples it has in the full dataset. The tested objective does not replace CE completely; it adds the metric-derived term to CE so that optimization still rewards calibrated class probabilities while also pushing the minibatch toward higher macro soft F1:

$$\mathcal{L}_{\mathrm{CE+F1}} = \mathcal{L}_{\mathrm{CE}} + \left(1 - \frac{1}{C}\sum_{c=1}^{C} \mathrm{F1}_{\mathrm{soft},c}\right). \tag{17}$$

The benchmark trains this loss with balanced sampling on the patch manifest. That design is important for interpretation: the loss tries to optimize a macro-style class metric, while the sampler ensures that tail classes occur often enough in minibatches for their soft-F1 terms to provide a regular training signal.

2.5.2 Scholz-Style CE + Soft MCC

The soft-MCC variant uses the same principle as soft F1—replace hard evaluation counts with differentiable probability masses—but starts from the Matthews correlation coefficient (MCC).

Whereas F1 for one class depends mainly on that class's positive predictions, MCC is built from the full confusion matrix and therefore reacts to false positives, false negatives, and true negatives at the same time. In binary problems it is often preferred in imbalanced settings because it can remain informative when one class dominates the dataset: accuracy can look high while predictions are still systematically biased, but MCC penalizes both kinds of mistakes jointly.

Binary MCC as a correlation coefficient. For one class treated as positive, write TP, TN, FP, and FN for the four confusion-matrix cells. The standard MCC is

$$\mathrm{MCC} = \frac{\mathrm{TP}\cdot\mathrm{TN} - \mathrm{FP}\cdot\mathrm{FN}}{\sqrt{(\mathrm{TP}+\mathrm{FP})(\mathrm{TP}+\mathrm{FN})(\mathrm{TN}+\mathrm{FP})(\mathrm{TN}+\mathrm{FN})}}. \tag{18}$$

The numerator compares co-occurrence of correct positive and correct negative decisions with co-occurrence of the two error types; the denominator scales by how many examples fall into each margin of the confusion matrix. MCC equals +1 for perfect predictions, 0 for chance-level association, and can become negative when predictions are worse than random. This makes it a correlation-style score rather than a prevalence-weighted rate.

Multiclass MCC in a minibatch. For C cancer types, the multiclass extension treats labels and predictions as C-dimensional vectors per example. Let $Y \in \{0,1\}^{N\times C}$ be the one-hot label matrix and $\widehat{Y} \in [0,1]^{N\times C}$ the corresponding softmax probabilities, with rows $\mathbf{y}_i$ and $\hat{\mathbf{y}}_i$. Three aggregate quantities describe how mass is distributed inside the batch:

$$s = \sum_{i=1}^{N}\sum_{c=1}^{C} Y_{ic}\widehat{Y}_{ic}, \qquad t_c = \sum_{i=1}^{N} Y_{ic}, \qquad p_c = \sum_{i=1}^{N} \widehat{Y}_{ic}. \tag{19}$$

Here s is the total probability-weighted agreement between labels and predictions: each example contributes $\hat{y}_{i,y_i}$, the predicted mass on its true class. The terms t_c and p_c are the batch totals of true and predicted mass for class c. They play the role of marginals in the confusion-matrix calculation. If predictions collapse onto a few head classes, the vector $(p_1, \ldots, p_C)$ becomes skewed even when argmax accuracy looks acceptable; MCC is sensitive to that mismatch because it compares the joint alignment s with what would be expected from the marginals alone.

The multiclass MCC used in training can be written as a normalized covariance between the label matrix and the prediction matrix:

$$\mathrm{MCC}_{\mathrm{mc}} = \frac{Ns - \sum_{c=1}^{C} t_c p_c}{\sqrt{N^2 - \sum_{c=1}^{C} p_c^2}\,\sqrt{N^2 - \sum_{c=1}^{C} t_c^2}}. \tag{20}$$

The numerator $Ns - \sum_c t_c p_c$ increases when true and predicted mass line up example by example and decreases when correct predictions are no better than what the marginal totals would suggest by chance. The two square-root terms measure how spread out the batch-level label totals and prediction totals are across classes; they prevent the score from exploding when one minibatch happens to contain only a few classes. Equation (20) reduces to the familiar binary form in (18) when $C = 2$.

Soft MCC loss. Hard MCC replaces $\widehat{Y}$ by one-hot argmax predictions, which again blocks gradient-based training. The soft-MCC objective keeps the structure of (20) but plugs in softmax probabilities directly:

$$\mathrm{MCC}_{\mathrm{soft}} = \frac{Ns - \sum_{c=1}^{C} t_c p_c}{\sqrt{N^2 - \sum_{c=1}^{C} p_c^2}\,\sqrt{N^2 - \sum_{c=1}^{C} t_c^2}}. \tag{21}$$

Every term in (19) and (21) is differentiable with respect to the logits that produce $\widehat{Y}$. A confident correct prediction increases s; probability placed on the wrong classes increases some p_c while leaving the matching t_c unchanged, which lowers the numerator; and overly peaked prediction vectors reduce the prediction-side denominator term, limiting the score even if a few examples are classified correctly. As with soft F1, the training objective turns a score to be maximized into a loss to be minimized and adds it to CE:

$$\mathcal{L}_{\mathrm{CE+MCC}} = \mathcal{L}_{\mathrm{CE}} + (1 - \mathrm{MCC}_{\mathrm{soft}})\,. \tag{22}$$

This variant is also evaluated only in the patch benchmark and is trained with class-balanced oversampling. The soft-F1 and soft-MCC variants therefore ask a focused question: once tail classes are regularly presented to the optimizer, does adding a differentiable version of a class-balanced evaluation metric improve the controlled patch classifier beyond CE and simple balanced sampling?

2.5.3 Odd-K-Out (OKO) Set Learning

Standard cross-entropy optimizes each example independently, which has two relevant consequences under class imbalance. First, the gradient signal for a tail class arrives only when that class is sampled, so rare examples can be overwhelmed by frequent ones. Second, hard-label cross-entropy encourages logit values to diverge without bound, which is a known driver of overconfidence and poor calibration (Guo et al., 2017). OKO (Muttenthaler et al., 2024) addresses both problems simultaneously by replacing per-example cross-entropy with a set-level objective. Calibration is inherently a property of a model's output distribution over a collection of examples, and training on sets is a natural way to expose the model to that distributional structure during learning.

Set construction. A training set S of size $k + 2$ is constructed as follows. A pair class y' is sampled uniformly from $[C]$, equalizing class frequency in expectation regardless of the training distribution—analogous to balanced sampling but applied at the set level rather than the sample level. Two examples x'_1, x'_2 are drawn from the class y', and k distinct odd classes $y'_3, \ldots, y'_{k+2}$ are sampled without replacement from $[C] \setminus \{y'\}$. One representative example x'_j is drawn from each odd class y'_j. The resulting set $S = (x'_1, \ldots, x'_{k+2})$ therefore always contains exactly one matched pair and k distractors from different classes.

Set-aggregated loss. For a model f_θ parameterized by θ, OKO defines the aggregated logit vector as the sum of individual per-example outputs over the set members:

$$f_\theta(S) = \sum_{i=1}^{k+2} f_\theta(x'_i). \tag{23}$$

Prediction on the aggregated logits uses ordinary softmax. The pair class y' appears twice in the set and its two logit contributions therefore reinforce each other, raising the aggregate score for y' relative to any single odd class. This pooling effectively averages the raw logit values across set members, which limits how extreme the aggregate can be—an implicit logit regularization that the paper links to improved calibration even when training with hard labels.

The main training loss predicts the pair class from the aggregated logits:

$$\ell_{\mathrm{OKO}} = -\log \mathrm{softmax}(f_\theta(S))_{y'}. \tag{24}$$

Following the main-text configuration of the original paper, the implementation also trains an auxiliary classification head g_θ that shares the trunk with f_θ and predicts the first odd class y'_3:

$$\ell_{\mathrm{aux}} = -\log \mathrm{softmax}(g_\theta(S))_{y'_3}. \tag{25}$$

The total per-set loss is $\ell_{\text{OKO}} + \ell_{\text{aux}}$. The auxiliary head is discarded at inference, and single-example evaluation proceeds exactly as for any other classifier: the model accepts one patch feature vector and returns the logits from the main head. There is no inference overhead or architecture change at test time.

OKO belongs in the hybrid sampling–loss category because the set construction couples a balanced class-sampling policy with a modified training objective; neither component is meaningful on its own. Balanced sampling via uniform pair-class selection ensures that all C classes contribute equally to the gradient signal in expectation, which directly addresses the data-level imbalance problem. The aggregated set-level loss then introduces an implicit regularization on logit magnitude. Together these are not separable in the same way that standard cross-entropy and a WeightedRandomSampler are separable. In this benchmark OKO is evaluated in the patch feature regime only; the set-construction mechanism does not extend naturally to the attention-MIL bag encoder used in the WSI-bag benchmark.

The number of odd classes k controls set size and therefore the degree of logit averaging, and it is the imbalance control selected on validation. The paper's ablation experiments show that accuracy and calibration are insensitive to the specific value of k, and the default of $k = 1$ is the computationally cheapest option; a moderate set size can nonetheless work best in practice without requiring the largest sets.

2.6 Representation and Architecture-Level Methods

Representation-level methods try to make rare classes more separable in feature space rather than only changing sampling or per-example loss weights. This family is especially relevant for computational pathology, where WSI labels supervise bags of many unlabeled instances and class imbalance can occur both across slides and within a positive slide (Juyal et al., 2024).

2.6.1 Classifier Re-Training (cRT)

Classifier re-training (cRT) separates representation learning from classifier fitting (Kang et al., 2020). Stage one trains the ordinary CE model on the observed imbalanced training distribution. Stage two freezes the learned representation, discards and reinitializes the final linear classifier, and fits only that classifier using class-balanced sampling. If $h_{\widehat{\theta}}(x)$ is the frozen representation learned in stage one, cRT estimates

$$(\widehat{W}, \widehat{\mathbf{b}}) = \arg\min_{W,\mathbf{b}} \mathbb{E}_{(x,y)\sim q_{\text{bal}}} \left[-\log \operatorname{softmax}\big(W h_{\widehat{\theta}}(x) + \mathbf{b}\big)_y \right], \tag{26}$$

where q_{bal} gives classes equal expected exposure. This isolates whether imbalance primarily distorts the decision boundary after a useful representation has already been learned.

In the patch regime, stage one learns the common MLP head on frozen Virchow2 embeddings; cRT freezes its learned hidden representation and retrains a reinitialized linear output layer with balanced patch sampling. In the WSI regime, stage one learns the instance projection and attention aggregator; cRT freezes both components and retrains only a reinitialized bag classifier with balanced slide sampling. Validation selects the stage-two checkpoint, and its compute is reported in addition to stage-one training. The original cRT study considers several classifier normalization variants; the benchmark uses only the reinitialized linear-classifier variant specified here.

2.6.2 SC-MIL

SC-MIL extends supervised contrastive learning to MIL by applying the contrastive objective to bag representations instead of assigning noisy slide labels to individual patches (Juyal et al., 2024). The motivation is that a slide label usually does not describe every patch in a bag.

A cancer-type label can be correct for the slide while many instances contain background, normal tissue, technical noise, or weakly informative morphology. Applying contrastive learning directly to instances would therefore treat many patches as positives or negatives using labels they do not truly carry. SC-MIL avoids that mismatch by contrasting the aggregated slide-level representation.

The attention-MIL backbone first maps bag B_i to an embedding $\mathbf{b}_i = \mathbf{b}(B_i)$. A projection head g then produces normalized contrastive features $\mathbf{r}_i = g(\mathbf{b}_i)$. In the contrastive view, each bag in a minibatch is an anchor. Bags with the same cancer-type label are positives because their slide-level representations should become closer, while bags from other classes act as negatives because their representations should remain separated. For anchor i, bags of the same class form the positive set $P(i)$, and the supervised contrastive term is

$$\mathcal{L}_{\mathrm{SCL}} = -\frac{1}{\sum_i |P(i)|} \sum_i \sum_{p \in P(i)} \log \frac{\exp(\mathbf{r}_i^\top \mathbf{r}_p / \tau)}{\sum_{a \neq i} \exp(\mathbf{r}_i^\top \mathbf{r}_a / \tau)}. \tag{27}$$

The numerator rewards similarity to same-class bags. The denominator compares that similarity against all other bags in the minibatch, so the loss favors representations where class clusters are compact and mutually separated. The temperature τ controls how sharply these similarities are weighted; the reference default is $\tau = 1.0$, and τ is the imbalance control selected on validation.

SC-MIL does not rely on contrastive learning alone. The contrastive term shapes the representation space, while CE trains the final class predictor. The tested variant combines them through a linear curriculum:

$$\mathcal{L}_t = \beta_t \mathcal{L}_{\mathrm{SCL}} + (1 - \beta_t)\mathcal{L}_{\mathrm{CE}}, \qquad \beta_t = 1 - \mathrm{progress}_t. \tag{28}$$

Early training therefore emphasizes class-structured bag embeddings, when the classifier head is still unstable. As training progresses, β_t decreases and more weight moves to CE, so the learned representation is tied back to the actual slide-level classification objective. Class-aware batching draws at least two slides for each represented class whenever class support permits; otherwise an anchor would have $|P(i)| = 0$ and contribute no contrastive term. Training logs the number of valid anchors, ordered positive pairs $\sum_i |P(i)|$, and represented classes per batch. The normalization in Equation (27) uses the actual number of ordered positive pairs, so unequal class multiplicities do not silently change the scale of the contrastive objective.

2.6.3 MDE-Inspired Dual-Expert MIL

The MDE-inspired variant uses distribution-specific experts and a cross-expert consistency term for long-tailed WSI classification (Ling et al., 2025). The full source method combines a dual-expert ensemble with multimodal distillation from a pathology foundation model. The benchmark implements only the ensemble component: a shared attention aggregator, two expert classification heads, and optional logit consistency under the same frozen Virchow2 bags as the other MIL methods. It omits CONCH-based multimodal distillation and is therefore not called a faithful MDE-MIL reproduction.

The shared trunk matches plain attention MIL: each bag $B = \{\mathbf{x}_1, \ldots, \mathbf{x}_m\}$ is encoded instance-wise, attention weights are computed with a linear scorer and softmax over instances, and the weighted sum yields a bag embedding $S(B) \in \mathbb{R}^d$. Two separate feed-forward experts then map S to logits:

$$E_U(S) = W_U\, \phi_U(S) + b_U, \qquad E_B(S) = W_B\, \phi_B(S) + b_B, \tag{29}$$

where ϕ_U and ϕ_B are two-layer MLPs with ReLU and dropout. Expert U is trained on the natural slide distribution and expert B on a class-balanced slide distribution.

Each training step draws one minibatch from each distribution. The unbalanced loader shuffles slide bags as in MIL CE; the balanced loader uses the same inverse-frequency weights as balanced-sampling MIL. For batches (B_U, y_U) and (B_B, y_B), the classification loss is

$$\mathcal{L}_{\text{cls}} = \text{CE}\big(E_U(S(B_U)), y_U\big) + \text{CE}\big(E_B(S(B_B)), y_B\big), \tag{30}$$

and the cross-expert consistency term penalizes disagreement on the *same* bag embedding:

$$\mathcal{L}_{\text{con}} = \text{MSE}\big(E_U(S(B_U)), E_B(S(B_U))\big) + \text{MSE}\big(E_B(S(B_B)), E_U(S(B_B))\big). \tag{31}$$

The total objective is $\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_{\text{con}}\mathcal{L}_{\text{con}}$. The reference default is $\lambda_{\text{con}} = 0.25$, matching the Camelyon+-LT default reported by Ling et al.; λ_{con} is selected on validation. Setting $\lambda_{\text{con}} = 0$ defines the prespecified *dual experts without consistency* component ablation: both natural and balanced experts remain, but only their classification losses are optimized. Unlike RankMix-inspired mixing, this method has no teacher or student reinitialization; both experts share the aggregator throughout training.

At validation and test time, each slide bag is evaluated once in natural order with mean-logit ensemble inference:

$$\hat{\mathbf{z}}(B) = \tfrac{1}{2}\big(E_U(S(B)) + E_B(S(B))\big). \tag{32}$$

MDE-inspired dual-expert MIL is categorized as a hybrid WSI method because it couples two sampling distributions with an optional consistency regularizer rather than changing only the sampler or only the per-example loss.

3 Summary

This report defines the benchmark's no-mitigation references (CE and MIL CE) and its retained interventions: balanced sampling, class-weighted CE, focal loss, train-time and post-hoc logit adjustment, soft-F1, soft-MCC, CFAL, OKO, RankMix-inspired feature mixing, cRT, SC-MIL, and MDE-inspired dual-expert MIL. Balanced Softmax is recorded as related prior-correction literature rather than an additional tested method. The definitions make the tested adaptations, output semantics, and ablations explicit; the companion experimental protocol defines how their discrimination, probability quality, and mitigation recovery are evaluated.

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